

speak: Improving Confidence in Public Speaking

Naisha Agarwal

Department of Computer Science
UCLA
Los Angeles, California, USA
naishaa@g.ucla.edu

Travis Nguyen

Department of Computer Science
UCLA
Los Angeles, California, USA
travisnguyen@ucla.edu

Cristobal Velazquez

Department of Computer Science
UCLA
Los Angeles, California, USA
crisvela@ucla.edu

Jenna Wang

Department of Computer Science
UCLA
Los Angeles, California, USA
jayenay@ucla.edu

Eunice Jun

UCLA
Los Angeles, California, USA
emjun@cs.ucla.edu

The screenshot displays the SPEAK interface. On the left is a video player showing a woman in a red shirt speaking in front of a whiteboard. Below the video player is a list of feedback points: 'give brief introduction to filler words.', 'give some examples of filler words.', and 'explain how pauses can be used as a replacement for filler words.' On the right, under the heading 'content analysis', there is a section for 'outline feedback' with a dropdown menu set to 'pros'. Below this, specific feedback is provided for two time points: '00:16: give brief introduction to filler words.' and '16:16: give some examples of filler words.' A 'None' option is also present. Further down, another dropdown menu is set to 'areas of improvement'. Below this, feedback for '47:04' is shown: 'explain how pauses can be used as a replacement for filler words.' This is followed by a paragraph of detailed feedback: 'This topic was touched upon briefly ("There's going to be filler space if you can fill it with silence") but could have been more thoroughly explained and expanded upon. Consider dedicating a section around 8-10 minutes in to explore this idea further, perhaps with examples or personal anecdotes.'

Figure 1: Overview of **SPEAK**'s feedback for a speech. **SPEAK** is a platform where users can practice their speeches and receive feedback on the content and delivery (e.g., body language).

Abstract

Anxiety from public speaking can stem from two potential sources: content and delivery. Not only can content be difficult for users to devise, but users may also have trouble ensuring the end result is appropriate in terms of depth, length, and tone. When time comes to rehearse, a second opinion may not be available, causing mistakes during practice to be carried over to the actual speech. Through our AI pipelines, our system delivers personalized content assessments

and rehearsal feedback. Initial user evaluations demonstrated that the system helps users sharpen their message and increase clarity of delivery.

CCS Concepts

• **Human-centered computing** → HCI design and evaluation methods; **Interactive systems and tools**.

Keywords

Public speaking, rehearsal feedback, generative AI, multimodal AI

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1 Introduction

Public speaking is an essential skill, but can oftentimes be challenging to master. Effective practice requires timely, constructive feedback, yet many individuals practice alone without someone to critique their content or delivery. Without external input, identifying issues such as unclear structure, filler words, or distracting body language becomes difficult.

To address this, we developed *SPEAK*, a platform designed to streamline the feedback process, making it more efficient and accessible. *SPEAK* enables users to practice delivering speeches and receive detailed feedback on their content and delivery. Through leveraging multi-modal generative AI models, *SPEAK* analyzes recorded speech videos that users either upload or record directly within the platform. A demo video of *SPEAK* in action is available at <https://youtu.be/Z2RjNliWy9o?si=loMKSEN9DXv3VWID>.

2 Design Process

We engaged in an iterative design process to develop *SPEAK*. To understand people’s experiences with public speaking, we conducted 25 semi-structured interviews and collected survey responses from over 50 people. Our interviews explored topics such as public speaking challenges, contexts, confidence, habits, and strategies for improvement.

In addition, an integral part of our design process involved engaging with the Santa Monica Toastmasters, an organization for professionals to practice public speaking and develop leadership skills. We attended one of their weekly meetings and learned about impromptu speaking exercises, prepared speeches, and detailed evaluations, with roles such as Grammarian and Filler Word Counter offering targeted feedback. These observations, along with informal interviews with Toastmasters members and officers, provided rich insights into effective training structures.

There were several key themes we derived from user research. Several participants expressed a desire for low-pressure, accessible ways to practice public speaking outside of formal or high-stakes environments. Many cited a lack of constructive feedback as a barrier to improvement, emphasizing the value of detailed, specific, and actionable input. Confidence and anxiety management also emerged as central concerns, with users seeking tools that could help them gradually build self-assurance over time. Based on these insights, we prioritized features such as flexible video recording, structured feedback on both content and delivery, and continuous progress tracking to support sustained skill development (see Figure 1).

3 System

SPEAK is a web-application whose front-end is built using React, TypeScript, Vite, and Tailwind CSS. The primary interaction methods with *SPEAK* include (i) recording or uploading a video of a speech practice and (ii) viewing analysis results.

There are two major dimensions of speech analysis in *SPEAK*: content and delivery. For content, *SPEAK* uses OpenAI’s Whisper model [1, 2] to generate a transcript from the video. From there, it uses Llama 3 to extract filler words and calculate speech rate. If the user provides a script or outline for their speech, *SPEAK* will compare the user’s rehearsal with the transcript. For delivery, *SPEAK* uses

Gemini 2.5 Pro, a model with multi-modal capabilities, including video processing. *SPEAK* passes the video recording of the speech with carefully crafted prompts to Gemini and receives a detailed analysis of a user’s body language at specific timestamps in the video. The timestamps are provided to the user so they can go back and refer to the highlighted points.

Each type of analysis uses a separate prompt, crafted through rounds of experimentation and assessment, and runs through their own dedicated AI pipeline for greater control. The analysis engine is built with a FastAPI server and a MongoDB database to store speech sessions.

4 Preliminary User Evaluation

We conducted a user study with twelve participants to assess the (i) usability and (ii) usefulness of *SPEAK*.

Users were evaluated in-person, in which they went through the process of recording a speech and receiving feedback from *SPEAK*. We gave users a set of impromptu topics to choose from and asked them to use *SPEAK* to record and analyze their speech. During this process, we primarily took note of participants’ actions, such as what they did and did not click and what they read. Some users would also walk us through what they were thinking by speaking aloud, such as points of confusion or improvement, which we noted as well. We then conducted a thematic analysis of our notes. We summarize key findings below.

4.1 Findings

Missing or Confusing Guidance. Participants expressed a desire for more guidance at key points in the experience. Users experienced confusion when choosing between content and delivery analysis. Many were unclear about whether the options were mutually exclusive and did not understand that both required video input. Some users were also unsure about the differences between input types, such as script versus outline, and did not immediately realize that elements like thumbnails or feedback boxes were clickable. These insights suggest an opportunity to improve clarity through more descriptive instructions, onboarding prompts, and clearer visual cues for interactivity.

Loading and Latency Issues. Users noted long wait times after recording or uploading a video, with delays in receiving feedback. The “analyzing video...” screen felt stagnant, and placeholder messages like “no delivery analysis” appeared before results loaded, causing uncertainty. Participants suggested adding animations or a progress indicator for better feedback.

Positive Feedback and Intuitiveness. Participants generally found the interface intuitive and user-friendly. Features such as clickable timestamps and accurate body language feedback were especially well-received, and users expressed appreciation for the overall experience.

5 Limitations and Future Work

We identified several limitations and areas for future development through our preliminary user evaluation. Our current system primarily supports speeches of duration up to five minutes. While *SPEAK* will work on longer speeches, the analysis it provides is

slower or less accurate. Because it was developed without group presentations in mind, providing definite support for it is also another consideration. Future improvements raise new questions and opportunities around how to prioritize actionable feedback within a collaborative context, especially for longer speeches.

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